

STAT 515 Final Project

The Impact of Student Habits on Academic Performance

GROUP 1:

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Abstract

This research examines the interrelationships between student behavioral patterns and academic outcomes through statistical modeling techniques contained within the program of instruction for Statistics 515 Applied Statistics and Visualization for Analytics. We analyzed a data set from Kaggle which was collected from 1,000 undergraduate students. Our research focused on three fundamental inquiries: (1) the extent to which student scores can be predicted from study habits and lifestyle factors (2) methodological improvements to prediction accuracy, and (3) the utility of classification models to identify students likely to achieve passing scores of 70% or higher. Our analysis revealed relationships between academic performance and certain behavioral indicators, specifically study hours, attendance, and screen time. Among regression models, Lasso regression demonstrated the best predictive capacity ($R^2=0.792$, $RMSE=7.557$). Stepwise regression confirmed that a minimal variable subset could retain sufficient predictive power while also enhancing interpretability.

Introduction

This research aims to identify the optimal models to utilize in determining the variables which affect academic outcomes most significantly. From cognitive abilities of students and traditional predictors, the research indicated other factors can also have a significant effect on academic outcomes. Our research adds to this area of study by examining the relationship between daily habits, lifestyle choices, and examination performance among undergraduate students. We used multiple regression and classification models to examine three research questions to derive insights of the major contributing factors of exam scores and whether a student passes or fails the exam.

Dataset Description

This research examines the student habits dataset which contains 15 variables and 1000 rows. The dataset contains a variety of student habits such as study hours, sleep hours, student part time job status, etc. This data comprises of detailed records of students in the undergraduate or graduate levels of studies along with their usual habits in and out of school.

This dataset was sourced from Kaggle.com and is synthetic data built upon the habits of students of this level. The dataset contained no missing values or bad formatting, however certain datatypes needed to be factorized in order to proceed with further use and analysis. Academic behavior is represented by study hours per day, attendance percentage, and extracurricular participation. Lifestyle factors are also considered in the dataset such as sleep hours, diet quality, exercise frequency, and hours spent on social media. Lastly, the demographic features of the student are represented by age, gender, and their parental education level. All of these factors are key to uncovering if there are trends in exam scores per variable or if there are certain variables that are more important than others in predicting the outcome of the exam.

Variable	Values
Age	17-24 years
Gender	Female, Male, Other
Parental education level	None, High School, Bachelor Master
Study hours per day	0-8 hours
social media hours	0-6 hours
Netflix hours	0-5 hours
Part-time job	Yes/No
Attendance percentage	60-100%
Sleep hours	3-10 hours
Diet quality	Poor, Fair, Good
Exercise frequency	0-6 times per week
Internet quality	Poor, Average, Good
Mental health rating	(1-10)
Extracurricular participation	Yes/No
Exam Score	~ 20-100

Research Questions

We have analyzed the data and built models that pertain to the questions of how certain variables affect exam scores/performance of the student. We chose three interconnected research questions which build upon one another. These questions can explore how claims of common ways to improve exam scores can be supported or disproved through data-backed evidence and future predictions by various machine learning models.

1. Can we predict student scores based on lifestyle, study habits, attendance, and sleep hours?
2. How (if at all) can predictive performance be improved from prior models?
3. What factors most significantly predict whether a student will pass an exam with a score of 70 or higher?

Data Cleaning and Organization

The dataset sourced from Kaggle.com, was mostly clean in CSV formatting. However, while it was clean, extra steps were taken in order to allow the data to be further analyzed. The categorical values were converted into factor values which makes model building more efficient and flexible. Factorization is important in this case because many statistical models require all input features to be numeric to even run. A requirement of feature importance is that all categories must be factorized and will enable most easy interoperability for regression and classification outputs. Feature engineering was utilized to create two variables, “screentime” and “pass”. Screentime is the sum of hours on social media and Netflix; whereas pass classifies the students who achieved an exam score of 70 or above as “pass” and below 70 as “fail”.

Data Exploration

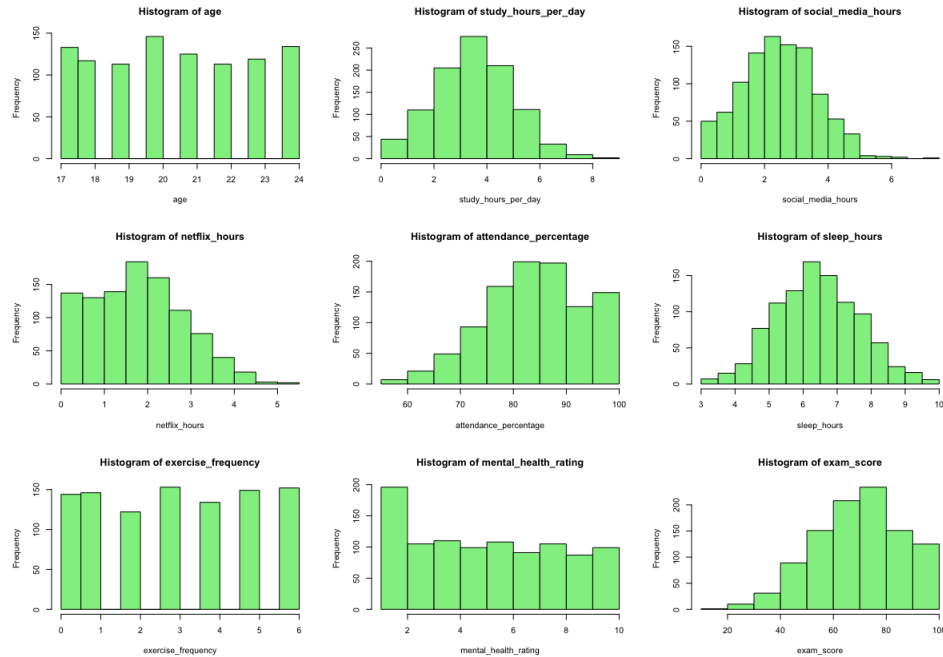
The analysis of the data revealed several specific characteristics which contribute to understanding of the sample:

- Age Distribution: students in the sample population ranged from 17 to 24 years old. There was a uniform distribution across the developmental spectrum.
- Gender distribution: there was a larger sample size of females (45.2%) as compared to males (42.9%), with a minimal population self-identifying as “other” (11.9%)
- Study hours: the daily academic engagement outside of class hours in general ranged from 0 to 8 hours with a mean of 4.2 hours.
- Attendance: Classroom participation rates were high with a mean of 85.3% and a minimum threshold of 60%.
- Exam Scores: Academic performance of the sample size ranged from 20 to 100 points with a mean of 70.2.

Our analysis of the variable distributions highlighted several significant patterns, which could be used for subsequent analysis:

- Study hours per day: the distribution was centered around 4 hours per day and there was a gradual decrease at the extremes, most significantly at the upper limits of the sample.
- Sleep hours: on a normal distribution that is centered around 6.7 hours. A small number of students reported sleep patterns below traditional physiological thresholds (<4 hours).
- Attendance percentages: the sample group had a high rate of attendance (90%), this indicates a norm favoring class participation and this could be reinforced based on academic grading policies which include attendance metrics.
- Exam score: analysis of the data revealed that the frequency of students' scores incrementally increased up to approximately 80 points and then the frequency of higher scores was significantly reduced after that point. From the dataset, scores peaking at 80 points highlight the mean value of the scoring across the population.

Figure 2



Our analysis indicated both positive and negative correlations between the attributes in the dataset. On the positive side and unsurprisingly, there appeared to be strong positive correlations between exam scores and study hours ($r=0.82$). Secondly, the attendance showed weak, but positive, correlation to mental health rating ($r=0.32$). Exercise frequency also showed weak correlation with exam scores ($r=0.16$, $r=0.12$), as did the amount of sleep hours ($r=0.12$). On the negative correlation side, screentime, not related to academic study, was a significant factor in lower exam scores. Based on figure 4, the boxplot represents the variables of study hours daily, exam score, and sleep hours. The boxplot indicates that there is a wide distribution spread of each of these variables such as study hours, one student may study for up to 10 hours a day while some other student may study for only 1 hour a day. From figure 4, the scatterplot displays that students who study more tend to have a higher exam score, there is a strong positive correlation between these two variables. Gender appears to be dispersed about equally, not showing a strong indicator that one gender has better academic performance than the other.

Figure 3

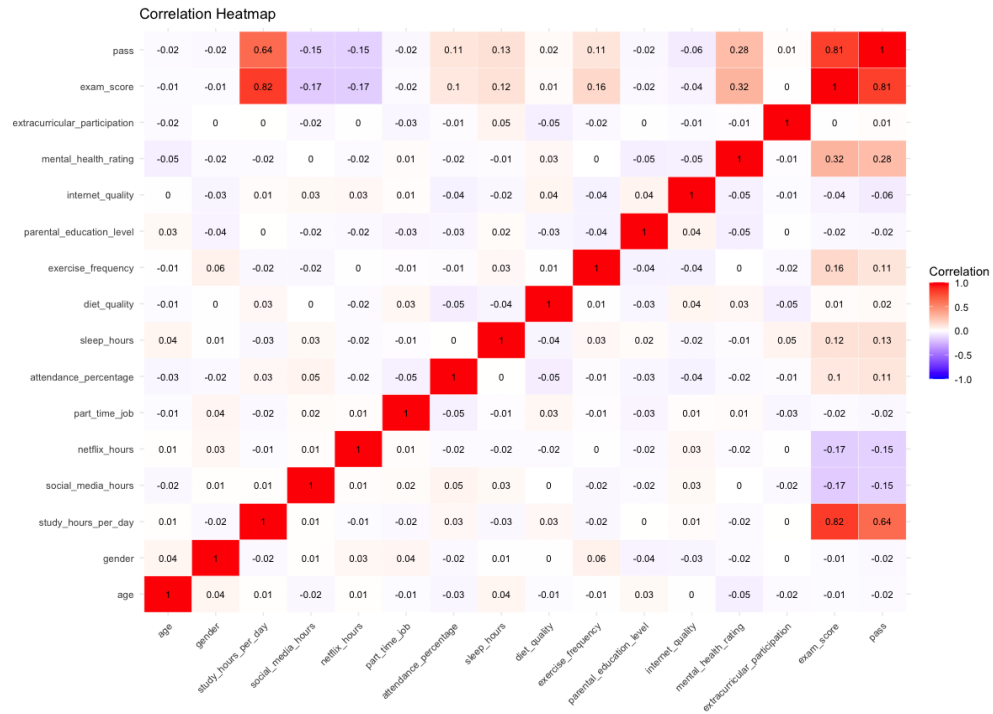


Figure 4

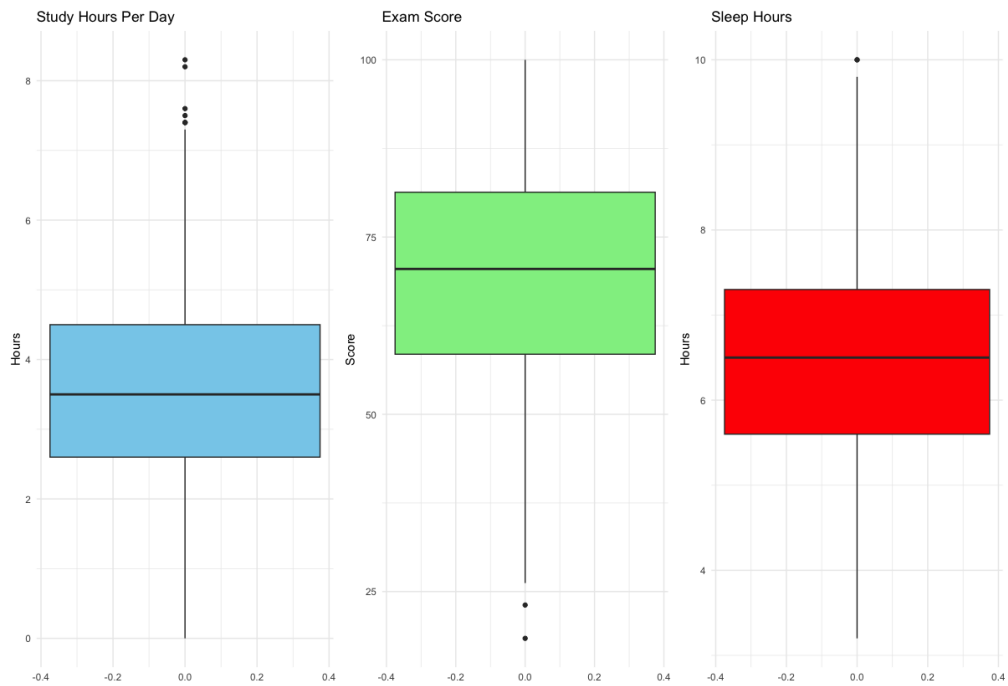


Figure 5



Methodology

Our analysis consisted of initially applying linear regression models to the dataset to help determine potential correlations related to the research questions. To address the first research question, a linear model was trained to an 80%-20% training to test partition of the dataset with the following predictors: study hours, sleep hours, part-time job, screentime, attendance, extracurricular participation, diet quality, exercise frequency as well as the interaction effect of study hours and sleep hours. A 10-fold cross-validation evaluation technique was applied during training to prevent overfitting. This model was evaluated using R-squared, RMSE and MAE metrics using the caret package.

To address the second research question, we trained Ridge, Lasso, and Random Forest models. Like the linear model, these models were trained, to an 80%-20% training to test partition of the dataset with the same predictors and evaluated using the same metrics. We later compared the results of the four models to determine which model performed the best.

For the third research question, we first needed to create a new variable, “Pass”, which classifies the exam scores less than 70% as “fail” and scores greater than or equal to 70% as “pass”. Afterwards, we ran a full logistic model with all predictors including the new predictor “Pass” and evaluated the significant predictors. A stepwise variable selection was then applied to select the most relevant predictors. To supplement our findings, an ANOVA statistical test was run to determine if there is any significant difference between the stepwise selected model and the full model.

Model Building

We first utilized the baseline linear model which resulted in identifying significant predictors of exam scores including study hours ($\beta = 9.62$), total sleep hours ($\beta = 2.02$), and screen time ($\beta = -2.50$). The model was able to explain approximately 78% of the variance in student performance.

For us to address potential statistical challenges, both Ridge and Lasso regression models were utilized. The Ridge regression model stabilized coefficient estimates while preserving all predictors. The Lasso model automatically removed less influential variables while achieving the overall best performance. Random Forest regression identified non-linear relationships but performed lower than Lasso.

To correctly conduct classification of passing or failing exam scores we utilized logistic regression to predict the pass or fail outcomes based off the exam scores (≥ 70). The logistic model provided an output of 89.25% accuracy with a sensitivity of 94.68% and a specificity of 85.00%. The stepwise selection process only retained the most significant predictors (study hours, screentime, attendance, sleep, exercise, and diet quality). The stepwise process removed gender, age, parental education level, extracurricular participation, internet quality, and part-time work status variables. A comparison of the stepwise model to the other models showed no statistical difference in performance ($p=0.6843$). An ANOVA hypothesis test was used to determine if the full model would be necessary over the reduced stepwise model.

Model Performance Comparison				
Model	R2	RMSE	MAE	Notes
Linear Regression (OLS)	0.789	7.793	6.339	Baseline Model
Ridge Regression	0.786	7.693	6.306	alpha = 0 lambda = 1
Lasso Regression	0.792	7.557	6.188	alpha = 0 lambda = 0.162
Random Forest	0.778	7.823	6.369	mtry = 4
Key Predictors Across Models				
Variable	OLS (β)	Ridge (β)	Lasso (β)	RF Importance (%)
Study hours per day	9.619	5.600	8.310	82
Total screentime	-2.502	-2.314	-2.405	100
Sleep hours	2.017	0.112	1.251	30
Attendance percentage	0.140	0.126	0.121	15
Exercise Frequency	1.509	1.370	1.418	62

Model Interpretation

Our analysis identified the Lasso regression model as the best model for predicting student exam scores. The study hours variable was the strongest positive predictor (coefficient ≈ 8.31 points per additional hour). The total screentime variable had the largest negative impact (coefficient ≈ -2.41 points per hour). There were several other significant positive variables which included attendance percentage (0.12 points per percentage point), sleep hours (1.25 points per hour), and exercise frequency (1.42 points per session). The variable for extracurricular activities was removed during the application of the Lasso model.

The application of the stepwise regression model to the dataset allowed for additional factors to be identified to delineate between passing and failing students. Like the Lasso model above, each additional hour of study time resulted in increased odds of passing and each increase percentage of attendance increased the likelihood of passing. In contrast, each hour of

screen time decreased the odds of passing. Sleep, exercise and a good quality diet increase the likelihood of passing by significant amounts.

Model Evaluation

The models used in this research included R^2 , RMSE, and MAE for regression analysis and accuracy, sensitivity, specificity, PPV, and NPV as our classification models. Through this process it was determined that the Lasso regression model performed the best at predicting exam scores (R^2 of 0.792, RMSE of 7.557, and MAE of 6.188). This was an improvement over the baseline linear regression model. The comprehensive logistic regression model was used for categorization as it outperformed the stepwise selected model, returning results with an accuracy of 89.25%. Using ANOVA to perform a hypothesis test for the full logistic regression model compared to a reduced stepwise model showed that though the full logistic model yielded better accuracy score, it is not by much (<1%) difference, we found that there is no statistically significant difference between the two models.

Performance Metrics on the Test Set

Model	R^2	RMSE	MAE
Linear Regression	0.789	7.793	6.339
Ridge Regression	0.786	7.693	6.306
Lasso Regression	0.792	7.557	6.188
Random Forest	0.778	7.823	6.369

Performance Metrics of Classification Models

Model	Accuracy	Sensitivity	Specificity	PPV	NPV
Comprehensive Logistic	89.25%	94.68%	85.00%	83.18%	95.33%
Stepwise	88.79%	94.68%	84.17%	82.41%	95.28%

Conclusion

The statistical analysis conducted on the student performance data set yielded several key findings. Most predominantly, the study hours and screentime variables were the best predictors of academic success. Each additional hour of study by a student resulted in an associated 8.31-point increase in exam scores. In contrast, screentime engagement, consisting of social media and Netflix, showed significant negative impacts on student performance with each hour of screentime linked to a 2.41-point decrease in scores.

This research yielded the following answers to our research questions.

- Can we predict student scores based on lifestyle habits, attendance, and sleep hours?

The Lasso model was able to achieve an R^2 of 0.792. This signifies that almost 79% of the variance in exam scores can be predicted by certain behavioral and lifestyle variables. The substantive variables positively contributing to student academic outcomes were study hours, attendance percentage, and sleep duration; while screentime contributed negatively.

- How (if at all) can predictive performance be improved from prior models?

In comparing the results of the trained linear, ridge, lasso and random forest models, it was revealed that the Lasso model produced improved performance. The lasso model was able to improve R^2 of the baseline linear model from 0.789 to 0.792 and reduce RMSE from 7.79 to 7.56 compared to conventional linear regression.

- What factors most significantly predict whether a student will pass an exam with a score of 70 or higher?

The comprehensive regression model was best suited for this question as it was able to discriminate among the population and determine to a high degree (89.25% accuracy) the students likely to pass. Though the comprehensive logistic regression model does better, it is not by much because the stepwise function has an accuracy of 88.79%. The full model may have a higher degree of accuracy; however, it is not by much. Hypothesis testing was done to address

this issue, and it indicates that there is no statistical significance that the full model is more efficient than the stepwise model.

Limitations and Further Research

This study has some limitations that have not been addressed in the study. Exploring other variables can create different outcomes, such as the inclusion of differing interactions and variables to see if accuracy of the models can be increased. Creating a new tailored model based on what the future study can allow for different perspectives on how the data and questions are approached. Exploring other types of advanced machine learning techniques such as neural networks can allow for more precise and accurate results. A major limitation of this study is the fact that the data worked on is synthetic, though based on some truth. Having a dataset using real surveyed data can allow for more accurate and scalable outcomes that can maybe be pushed to the public.

References

[1]

Naath, J(n.d.). *Student Habits vs Academic Performance* [Data set]. Kaggle.

<https://www.kaggle.com/datasets/jayaantanaath/student-habits-vs-academic-performance/data>